

Problem Statement

Contrastive Learning for Forensic Image Authentication

1 Aim and Representation Space

Let $\mathcal{X}$ be our dataset consisting of pristine images, where $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M \in \mathcal{X}$. Each image is defined in the spatial domain as $\mathbf{x}_i \in \mathbb{R}^{H_o \times W_o \times C_o}$.

Our objective is to learn a low-dimensional semantic representation that robustly captures the global spatial and structural features of the image, while remaining highly sensitive to localized, mathematically defined tampering. We define our representation function (encoder) as:

$$f_\theta : \mathbb{R}^{H_o \times W_o \times C_o} \longrightarrow \mathbb{R}^d$$

such that $d \ll H_o \times W_o \times C_o$. The representation of an image $\mathbf{x}_i$ is thus:

$$\mathbf{h}_i = f_\theta(\mathbf{x}_i) \quad (1)$$

where each $\mathbf{h}_i \in \mathcal{H}$, denoting the representation hypersphere.

2 Augmentation Framework ($\mathcal{T}$)

We restrict our augmentation space to mathematically formulated image degradation, specifically blurring and additive noise. We classify these into two distinct subsets: Global (Benign) and Local (Malicious).

2.1 Global Augmentations (Positive Pairs)

Let $\mathcal{T}_{global}$ denote transformations we consider semantically benign (e.g., standard transmission compression or camera defocus).

$$t_i(\mathbf{x}) = \mathbf{b}_i * \mathbf{x} + \mathbf{n}_i \quad (2)$$

where $t_i \in \mathcal{T}_{global}$, $\mathbf{b}_i \in \mathbb{R}^{K \times K}$ is a globally applied convolution kernel, and $\mathbf{n}_i \in \mathbb{R}^{H \times W \times C}$ represents i.i.d. noise distributed as $\mathcal{N}(0, \sigma^2)$.

Rationale: Positive pairs are generated from transforms rather than distinct images because if the model successfully pushes the representations of allowed transformations closer together, it proves the encoder has extracted the invariant, true semantics of the image regardless of benign degradation.

2.2 Localized Augmentations (Hard Negatives)

Let $\mathcal{T}_{local}$ denote malicious, localized tampering (e.g., concealing a watermark).

$$t_{loc}(\mathbf{x}) = (\mathbf{m} \odot (\mathbf{b}_{loc} * \mathbf{x})) + ((1 - \mathbf{m}) \odot \mathbf{x}) \quad (3)$$

where $\mathbf{m} \in \{0, 1\}^{H \times W}$ is a spatial binary mask defining the localized region, and $\odot$ is the Hadamard product. The encoder must explicitly isolate these anomalies and push their representations away from the pristine origin.

3 Hybrid Encoder Architecture (f_θ)

To satisfy the dual requirements of extracting high-frequency local textures and detecting global spatial anomalies, the encoder f_θ is designed as a Convolutional-Transformer Hybrid.

1. **Convolutional Backbone (f_{CNN}):** Natively processes the convolution operation of the blur kernels. It acts as a local texture extractor, mapping $\mathbf{x}_i \rightarrow \mathbf{F}_i$, where $\mathbf{F}_i \in \mathbb{R}^{H' \times W' \times C'}$ is a downsampled feature map retaining spatial topography.

2. **Patch Flattening:** The spatial grid is unrolled into a sequence of tokens. A learnable global [CLS] token, $\mathbf{z}_{cls}$, is prepended to the sequence.
3. **Transformer Head (f_{Tx}):** Applies Multi-Head Self-Attention (MHSA) to the token sequence. The attention mechanism compares the pristine regions against the structurally anomalous patches generated by $t_{loc}(\mathbf{x})$.

4 Mathematical Formulation of the Hybrid Encoder

The hybrid encoder processes the input image through a sequential pipeline designed to extract local high-frequency textures before evaluating global spatial anomalies.

4.1 Convolutional Feature Extraction (f_{CNN})

The image is passed through a truncated ResNet-50 backbone. ResNet-50 utilizes "Bottleneck" residual blocks, where the skip connection bypasses a sequence of three convolutional layers (1×1 , 3×3 , 1×1). Let $\mathbf{x}^{(l)}$ be the input tensor to the l -th residual block. The output tensor $\mathbf{x}^{(l+1)}$ is computed as:

$$\mathbf{x}^{(l+1)} = \text{ReLU}(\mathbf{x}^{(l)} + \mathcal{F}(\mathbf{x}^{(l)}, \{\mathbf{W}_k^{(l)}\})) \quad (4)$$

where $\mathcal{F}$ represents the residual mapping learned by the three convolution operations, batch normalization, and intermediate ReLUs. The final spatial feature map is $\mathbf{F} \in \mathbb{R}^{32 \times 32 \times 1024}$.

4.2 Tokenization and Projection

The spatial grid of $\mathbf{F}$ is flattened into a 1D sequence of N patches. These patches are linearly projected into the Transformer's hidden dimension D . A learnable classification token, $\mathbf{z}_{cls}$, and positional embeddings $\mathbf{E}_{pos}$ are added:

$$\mathbf{Z}_0 = [\mathbf{z}_{cls}; \mathbf{Z}_{patches}] + \mathbf{E}_{pos} \quad (5)$$

4.3 Transformer Head (f_{Tx})

The core mechanism is Multi-Head Self-Attention (MHSA), computing the interaction between all spatial patches:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (6)$$

If a localized region is tampered with, its high-frequency token features mathematically contradict the pristine context, driving the dot product $\mathbf{Q}\mathbf{K}^T$ toward zero and severing the attention score.

4.4 Projection to Representation Space

The final state of the [CLS] token forms our definitive semantic representation (the secure hash):

$$\mathbf{h}_i = f_\theta(\mathbf{x}_i) = \mathbf{z}_{cls}^{(L)} \in \mathbb{R}^{2048} \quad (7)$$

To compute the contrastive loss during training, we apply a non-linear projection head $g_\phi(\cdot)$ (a 2-layer MLP) to map the representation to a lower-dimensional space:

$$\mathbf{z}_i = g_\phi(\mathbf{h}_i) = \mathbf{W}^{(2)} \text{ReLU}(\mathbf{W}^{(1)} \mathbf{h}_i) \in \mathbb{R}^{128} \quad (8)$$

5 Modified NT-Xent Loss with Hard Negative Mining

5.1 Batch Generation and Similarity

For a random mini-batch of N images, we apply global augmentations to generate $2N$ views. For any image k , its augmented views $(\tilde{\mathbf{x}}_{2k-1}, \tilde{\mathbf{x}}_{2k})$ form a positive pair. The remaining $2(N-1)$ augmented images serve as standard negatives.

The similarity between any two projected vectors is defined via the normalized dot product:

$$\text{sim}(\mathbf{z}_i, \mathbf{z}_j) = \frac{\mathbf{z}_i^T \mathbf{z}_j}{\|\mathbf{z}_i\| \cdot \|\mathbf{z}_j\|} \quad (9)$$

To control softmax sharpness, we introduce a temperature scalar $\tau > 0$, defining $s_{i,j} = \text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau$.

5.2 Objective Function (Old)

Standard Contrastive Learning (NT-Xent) forces the model to ignore augmentations. To sensitize the model to localized forgeries, we introduce the tampered image $\mathbf{x}_{\text{tampered}} = t_{\text{loc}}(\mathbf{x}_i)$ into the denominator of the loss function as a heavily weighted Hard Negative. Let $\mathbf{z}_{\text{tampered}} = g_\phi(f_\theta(\mathbf{x}_{\text{tampered}}))$.

The modified probability that j is the correct match for i , accounting for the hard negative, is defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}{\sum_{k=1, k \neq i}^{2N} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau) + \alpha \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_{\text{tampered}})/\tau)} \quad (10)$$

where $\alpha \gg 1$ is the penalty weight assigned to the localized forgery. Because the denominator evaluates the negatives relative to anchor i , it is clear that $\ell_{i,j} \neq \ell_{j,i}$.

The final batch loss sums both directions for all positive pairs:

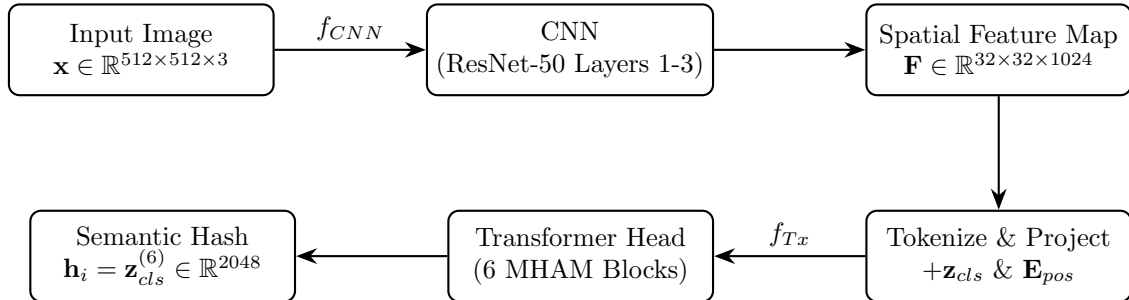
$$L = \frac{1}{2N} \sum_{k=1}^N (\ell_{2k-1, 2k} + \ell_{2k, 2k-1}) \quad (11)$$

6 Mathematical Signal Flow & Encoder Architecture

To satisfy the dual requirements of extracting high-frequency local textures and detecting global spatial anomalies, the encoder f_θ is designed as a Convolutional-Transformer Hybrid.

6.1 Signal Flow Graph

The following flow graph illustrates the transformation of the tensor from raw image space to the final semantic representation space:



6.2 Architecture Justification and [CLS] Interaction

The architecture isolates feature extraction from spatial reasoning to maximize forensic sensitivity:

1. **CNN Feature Extraction:** Standard Vision Transformers lack local inductive biases. By utilizing a truncated CNN backbone (f_{CNN}), we natively compute spatial gradients and extract localized high-frequency texture embeddings. A 512×512 image is downsampled into a grid of 1,024 spatial cells, where each cell inherently captures the presence or absence of local blur kernels ($\mathbf{b}_i$).
2. **Tokenization:** The spatial grid $\mathbf{F}$ is flattened into a sequence of 1,024 patch tokens and linearly projected to a hidden dimension of $D = 768$. A learnable global [CLS] token ($\mathbf{z}_{cls}$) and positional embeddings ($\mathbf{E}_{pos}$) are prepended to retain spatial topography.

3. **MHAM and Spatial Anomaly Detection:** During the Multi-Head Attention Mechanism (MHAM), the [CLS] token queries every spatial patch via the dot product $(\mathbf{Q}_{cls} \mathbf{K}_{patches}^T)$. If a localized region has been maliciously blurred, its low-frequency token features mathematically contradict the pristine global context of the surrounding patches. This misalignment drives the dot product toward zero, severing the attention weight. The [CLS] token absorbs this structural anomaly into its final state, yielding a highly sensitive, 2048-dimensional representation $\mathbf{h}_i$.

7 Asymmetric NT-Xent Loss with Hard Negative Anchoring

Standard Contrastive Learning (NT-Xent) generates positive pairs from two distinct augmentations of an image. However, in forensic applications, our localized tamper (Hard Negative) shares a massive pixel overlap with the pristine image. If a positive pair consists of two heavily blurred augmentations, the model may inadvertently assign a higher similarity score to the localized forgery than to the positive pair, leading to optimization collapse.

To stabilize training, we modify the objective to use **Asymmetric Anchoring**.

7.1 Pair Definitions

For a mini-batch of N pristine original images ($\mathbf{x}_{orig}$):

- **Positive Pair:** The pristine image is anchored to its globally blurred, semantically benign augmentation: $(\mathbf{x}_{orig}, \mathbf{x}_{global})$.
- **Hard Negative:** The spatially masked, locally blurred forgery: $\mathbf{x}_{tampered}$.

Let $\mathbf{z} = g_\phi(\mathbf{h}) \in \mathbb{R}^{128}$ represent the output of the MLP projection head. The similarity is defined by the temperature-scaled normalized dot product: $s(a, b) = \text{sim}(\mathbf{z}_a, \mathbf{z}_b)/\tau$.

7.2 Modified Objective Function

We calculate the loss by explicitly pulling the pristine anchor toward its global augmentation, while severely penalizing its proximity to the localized forgery. The asymmetric loss for image i is defined as:

$$\ell_i = -\log \frac{\exp(s(\mathbf{x}_{orig}^{(i)}, \mathbf{x}_{global}^{(i)}))}{\sum_{k=1}^{2N} \exp(s(\mathbf{x}_{orig}^{(i)}, \mathbf{x}_{global}^{(k)})) + \alpha \exp(s(\mathbf{x}_{orig}^{(i)}, \mathbf{x}_{tampered}^{(i)})} \quad (12)$$

where $\alpha \gg 1$ acts as the penalty multiplier for the localized anomaly. The denominator evaluates all standard negatives across the batch alongside the explicit hard negative. This formulation forces the optimization landscape to mathematically prioritize the detection of localized inconsistencies directly relative to the pristine anchor.